AAMA

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Fenestration & Glazing Industry Alliance

AAMA 501.2-25

Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls and Sloped Glazing Systems

an FGIA document

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AAMA 501.2-25, an FGIA Standard
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0.0 INTRODUCTION

0.1 The purpose of this document is to provide a quality assurance and diagnostic field water penetration check for fenestration product types specified within Section 1.0.

0.2 This specification establishes the requirements for field test specimens, apparatus, sampling, test procedures and test reports to be used in this diagnostic field water penetration check.

1.0 SCOPE

1.1 The procedure outlined in this document is not intended to test the rated or specified water performance representative of a wind driven rain event. AAMA 503, *Voluntary Specification for Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems*, is the proper test method for field testing of storefronts, curtain walls and slope glazing for air leakage resistance and water penetration resistance performance.

1.2 The procedure outlined in this document is not intended to test operable components such as operable fenestration products. AAMA 502, *Voluntary Specification for Field Testing of Newly Installed Fenestration Products,* is the proper test method for field air leakage resistance and water penetration resistance testing of newly installed operable windows and doors, and AAMA 511, *Voluntary Guideline for Forensic Water Penetration Testing of Fenestration Products*, is the proper test method for forensic water penetration testing of fenestration products.

1.3 This field check procedure is intended to evaluate those joints, gaskets and sealant details in the fenestration product which are designed to remain permanently closed and watertight and are not sealed using weatherstrip or weatherseals.

1.4 Products in scope

a) Curtain walls
b) Storefronts
c) Sloped glazing
d) Fixed windows and doors that do not incorporate a sash or panel
e) Fixed unit skylights that do not incorporate weatherstrip/weatherseals
f) Wall panels
g) Perimeter sealant
h) Air and water barrier integration with the fenestration product

1.5 Products out of scope

a) Operable fenestration products
b) Fixed windows and doors that incorporate a sash or panel
c) Entrance systems
d) Integral ventilating systems/devices
e) Operable unit skylights and roof windows

1.6 In this specification, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the specification; “shall be permitted to be” is used to express an option or that which is permissible within the limits of the specification; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express possibility or capability. Notes accompanying sections do not include requirements or alternative requirements; the purpose of a note accompanying a section is to separate explanatory or informative material from the text.

Notes to tables and figures are considered part of the table or figure and shall be permitted to be written as requirements. While Section 0.0 and any Notes not attached to figures and tables are non-mandatory, such notes provide important guidance , interpretation and clarification of the subjects noted.

1.7 The primary units of measure in this document are metric. The values stated in SI units are to be regarded as the standard. The values given in parentheses are for reference only.

1.8 This document was developed in an open and consensus process and is maintained by representative members of FGIA as advisory information.

2.0 SIGNIFICANCE AND USE

2.1 This document provides a method to check non-operable fenestration products and their installation during construction for water penetration resistance under controllable, reproducible, and appropriate conditions.

2.2 Improper use of this document may lead to a misunderstanding of a fenestration product's performance and/or the misapplication of fenestration products on the project.

2.3 This method has proven to be useful to manufacturers, installers, and other stakeholders as one part of a larger quality assurance or building commissioning program. However, when used for compliance with contract documents, the evaluation shall be performed by an AAMA accredited laboratory or field test agency.

NOTE 1: *Using this diagnostic check in isolation without performing water penetration resistance testing per AAMA 501.1, 502, 503, or 511 may not identify other potential leak paths in or around the fenestration product.*

2.4 The use of this document for the testing of installed fenestration products presumes that all personnel comply with the requirements of AAMA LAP-3, *Laboratory Accreditation Program Operations Manual- Laboratories and Test Agencies Performing On-site Testing of Fenestration Products*. These requirements include, but are not limited to, a working knowledge of the following:

- Building construction and fenestration product installation techniques.
- The various fenestration product types and designs, their operation and application.
- Isolation and systematic assessment of leakage through system components.
- The effect of interior and exterior ambient conditions (temperature, wind, barometric pressure, and moisture) on the specimen being tested.

2.5 All testing shall be conducted by an AAMA accredited laboratory or field test agency. See AAMA LAP-3 for more details.

3.0 REFERENCED DOCUMENTS

3.1 References to the standards listed shall be to the edition indicated. Any undated reference to a code or standard appearing in this section shall be interpreted as referring to the latest edition of that code or standard.

3.2 AAMA, Fenestration and Glazing Industry Alliance (FGIA) Standards

AAMA LAP-3, Laboratory Accreditation Program Operations Manual- Laboratories and Test Agencies Performing On-site Testing of Fenestration Products

AAMA 502-21, Voluntary Specification for Field Testing of Newly Installed Fenestration Products

AAMA 503-23, Voluntary Specification for Field Testing of Newly Installed Storefronts, Curtain Walls and Sloped Glazing Systems

AAMA 511-22, Voluntary Guideline for Forensic Water Penetration Testing of Fenestration Products

FG-24, FGIA Glossary

4.0 DEFINITIONS

Please refer to the most current FGIA Glossary for all definitions.

5.0 PREPARATION OF TEST AREA

5.1 All framing and/or units to be tested should be located on two typical floors of the building for curtain wall testing and on a typical floor for storefront or sloped glazing system testing.

5.2 The area shall be fully glazed to provide a completed installation. All of this work shall be done in strict accord with approved shop drawings and job specifications.

5.3 The architect and/or owner's representative shall designate an area of the completed portion to be tested. The test area shall be a representative sample of typical construction and shall have no outstanding punch list items or other visible defects. If no test area and/or location has been identified, the persons doing the test shall select an area.

5.4 This area shall be selected to provide representative information, usually a minimum of 9.3 m^2 (100 ft^2). The area to be tested shall include perimeter sealant, typical splices, frame intersections, and, if applicable, at least two entire vision lites and two entire spandrel lites containing an intermediate vertical member and an intermediate horizontal member.

5.5 This test is not intended for use on operable components such as windows and doors, Therefore, all operable components within the test area shall be protected and/or isolated from the water spray and are not within the scope of this field check. See Figures 1, 2 and 3 for typical test assemblies.

5.6 Testing shall be performed as soon as possible after the fenestration product is installed, sealants have been allowed to cure, and prior to the installation of drywall or other interior finish wall materials.

5.7 If interior finish wall materials have been installed, they shall be removed at the test area to allow visual access to check for water penetration, or other means of visual access shall be provided. If removal of interior finish work is required, the sponsor of the test is responsible for removal and repairs.

NOTE 2: *Interior finish materials do not include water sealing features integral to product performance. Each system manufacturer should have definitive installation instructions that identify ALL critical seals to be tested to achieve a performance rating. This information should be provided prior to testing.*

5.8 The test specimen shall be representative of typical installations and construction for the project. The specimen(s) shall have no outstanding punch list items or visible damage or irregularities, nor be singled out because of obvious performance problems.

5.8.1 Unless otherwise specified, exterior cladding, rain screen and associated components, including the exterior perimeter sealants, shall be installed within and/or adjacent to the fenestration product to represent the final detailed conditions of the exterior envelope. Allowance for the required curing times of materials and products adjacent to the test specimen shall be coordinated prior to testing.

5.8.1.1 If it is desired to test the air and vapor barrier and seals adjacent to the fenestration product prior to the exterior wall cladding being installed, the fenestration specimen perimeters shall be isolated during that testing and the fenestration product shall not be evaluated. The same isolation could be used to simulate the cladding and perimeter seals, should subsequent testing of the fenestration product be desired.

FIGURE 1: Exterior Wall Cladding and Associated Components Applied

NOTE 3: *Wind and other environmental conditions (barometric pressure, rain and temperature changes) can have an adverse impact on field testing performance; therefore, the specimen should be isolated from adverse ambient conditions with a barrier, and conditioned, if necessary, to protect it from any environmental condition which may affect the test results. If testing must be performed under conditions that would affect the test results, such as extreme temperatures, high winds or other adverse environmental conditions, these conditions should be discussed prior to testing and noted in the report. Performance of any test without appropriate consideration and adjustment for ambient conditions may render the test results invalid.*

NOTE 4: *Refer to AAMA 511 for the applicability of this field check to isolation testing for the determination of leakage paths.*

6.0 APPARATUS

6.1 The test shall be conducted using a Type B-25, #6.030 brass nozzle with a 13 mm (0.5 in) FPT as manufactured by Monarch Manufacturing Works, 7249-B Browning Road, Pennsauken, NJ 08109 (Phone: 856-241-1500).

6.2 The nozzle shall be connected to a hose [19 mm (0.75 in) diameter is suggested] and shall be provided with a control valve and a pressure gauge between the valve and the nozzle. The water pressure to the nozzle shall be adjusted to produce 205 to 240 kPa (30 to 35 psi) at the nozzle inlet.

6.3 The pressure gauge shall be calibrated at a maximum of six-month intervals. Records of calibration certificates shall be made available upon request.

6.4 If the water source is not capable of achieving calibration pressure, appropriate steps shall be taken to increase the source pressure. If the water pressure cannot be increased to the pressure specified in Section 6.2, it shall be noted in the test report along with the reason that the water pressure could not be achieved and that testing using a water pressure lower than the calibrated water pressure is a deviation from the field check.

7.0 PROCEDURE

7.1 The designated test area shall be divided into and evaluated in 1.5 m (5 ft) sections of the framing and joint. The nozzle shall be held normal to the plane of the wall and at a distance of 305 ± 25 mm (1 ft ± 1 in) from the location under test. Each 1.5 m (5 ft) section of test area shall be evaluated for a period of five minutes by slowly moving the nozzle back and forth over the test section (see Figure 4) while maintaining the nozzle perpendicular to the plane of the wall.

If the length of the section is less than 1.5 m (5 ft), then the section shall be tested for a duration equal to 1 minute per foot of section.

NOTE 5: *It is recommended that a gauge be attached to the end of the nozzle to ensure that the specified distance from the joint under test is maintained. The intent of the procedure is to indicate that each 1.5 m (5 ft) component (horizontal and vertical) is subjected to the water spray at a rate of 305 mm (1 ft) per minute. Therefore, it requires five minutes to test each 1.5 m (5 ft) length.*

7.2 Working from the exterior, the wall test section shall be selectively wetted progressing from the lowest horizontal framing member, then the adjacent framing intersections, then the adjacent vertical framing members, etc. During the test, an observer on the indoor side of the wall, using a flashlight, water sensitive paper, borescopes, mirrors, and other tools if necessary, shall check for any water leakage and shall note where it occurs.

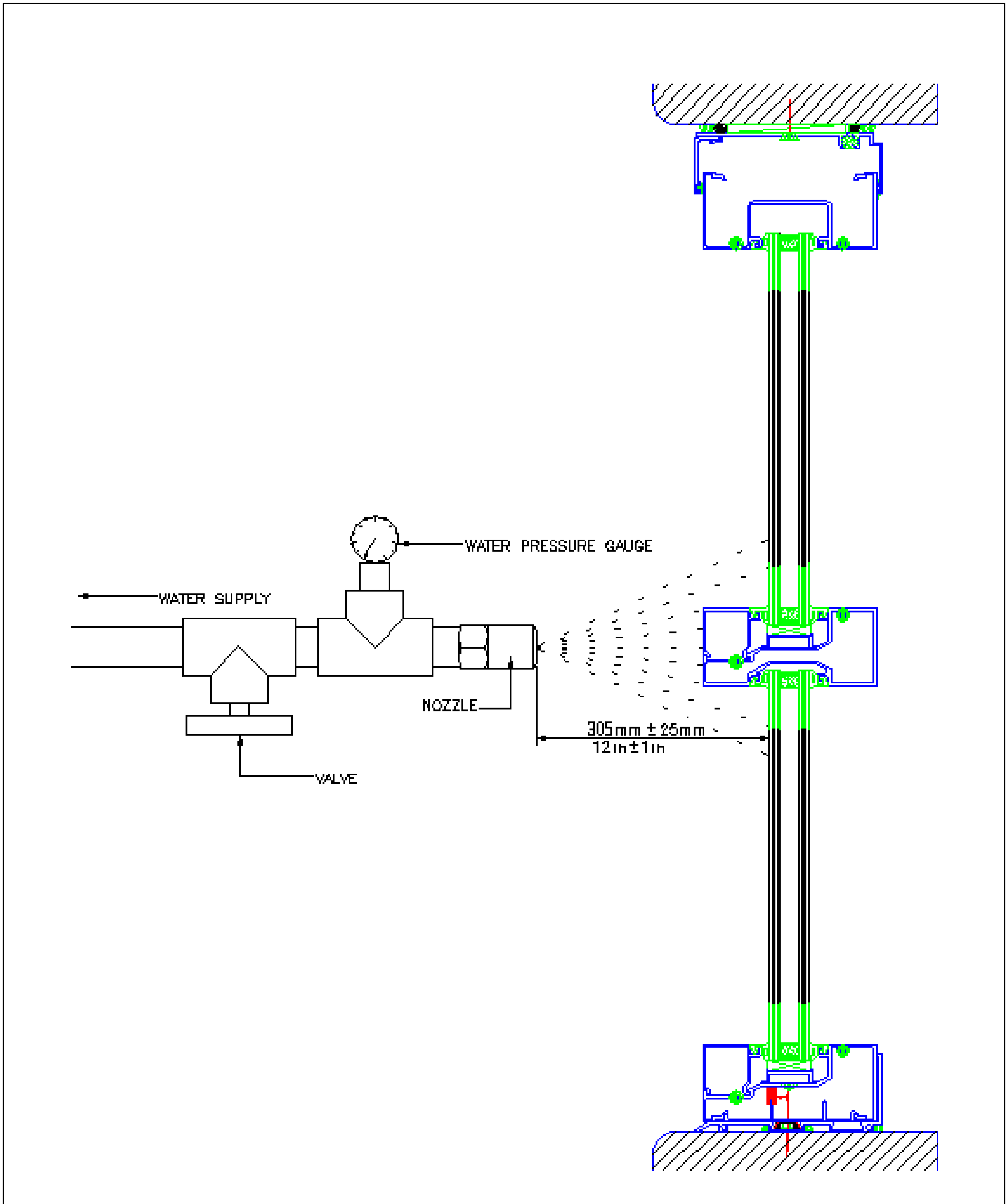

FIGURE 2: Water Leakage Field Check

7.3 For this water leakage field check, unless otherwise specified, water leakage is defined as any uncontrolled water that is not contained or drained back to the exterior, or that can cause damage to adjacent materials or finishes. Water contained within drained flashings, gutters, and sills is not considered water leakage. The collection of up to 15 ml (1/2 oz) of water in a five-minute test period on top of an interior stop or stool integral with the system is allowed. See Figures 3 and 4 for reference.

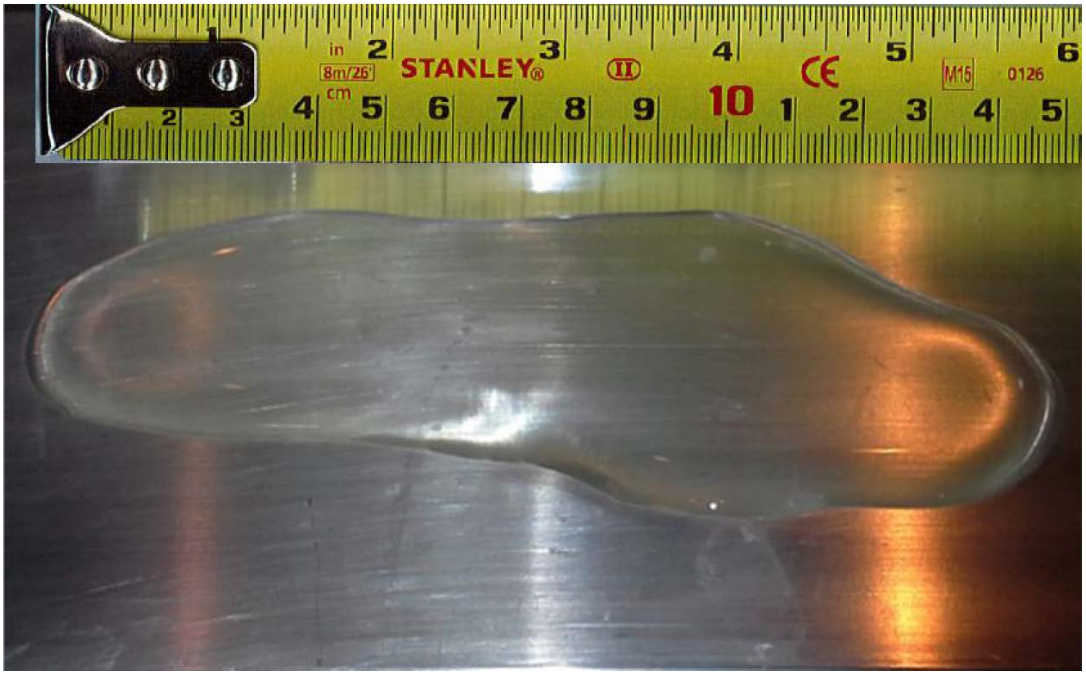

FIGURE 3: Example #1 of the Accumulation of 15 ml (1/2 oz) of Water

Figure 4: Example #2 of the Accumulation of 15 ml (1/2 oz) of Water

NOTE 6: *There are three typical paths for water leakage as presented AAMA 503: water penetration attributable to the surrounding wall conditions, water penetration attributable to the fenestration product specimen, and water penetration attributable to the perimeter joint.*

7.4 If no water leakage occurs during the five-minute test, the next 1.5 m (5 ft) of framing shall be wetted for five minutes, and testing continued in this manner until the entire test area is tested.

7.5 If water leakage occurs and the source of the leakage cannot be identified, the following sequence shall be followed:

7.5.1 After allowing the wall to dry completely and working downward from the top of the area to be checked, all joints, gaskets and framing within this area shall be completely and tightly covered on the outdoor side with a waterproof adhesive masking tape. If necessary, use small amounts of sealant where the tape wraps around the framing corners and joints to ensure that the masking is complete and waterproof.

7.5.2 Starting at the bottom of the prepared area, the masking tape shall be removed from the lowest horizontal framing for a distance of not more than 1.5 m (5 ft) from one end of the frame, including the joint intersection or corner at the end. This exposed length shall be subjected to the nozzle spray as directed in Section 7.1.

7.5.3 If no water leakage occurs during the five-minute test period, this length of framing shall be considered satisfactory and shall remain uncovered. If leakage has occurred at any point, the framing shall be re-taped at such points to prevent further leakage of these points during the subsequent checking of joints and framing adjacent to or above them.

7.5.4 This process shall then be repeated on all framing, gaskets and joint intersections within the designated area, using increments of exposed framing length not exceeding 1.5 m (5 ft) and always working upward on the wall.

NOTE 7: *In some cases, due to unforeseen delays or other causes, more than one working day may be required to completely check the designated area, necessitating that some or all of the masking tape be left in place overnight. The tape should not remain on finished metal or glass surfaces any longer than necessary, especially where subjected to strong sunlight, as this may make its removal difficult. It is the responsibility of the party contracting for the test to properly clean any residual adhesive or sealant resulting from the testing.*

7.6 When testing for compliance to contract documents, testing shall be performed by an AAMA accredited laboratory or field test agency.

NOTE 8: *AAMA accredited laboratories and field test agencies are required to comply with LAP-3. The requirement of AAMA accreditation assures the specifier that the laboratory or field test agency has the staff, training, experience and calibrated equipment to properly perform field testing.*

8.0 REMEDIAL WORK AND RE-CHECKING

8.1 If water leakage occurs as defined in Section 7.3, the manufacturer and/or the installer shall be afforded the opportunity to perform a site inspection and determine the reason for non-compliance. Non-compliant specimen(s) shall be repaired as required and retested as soon as practical**.** The remedial work shall be recorded and approved by the specifying authority, architect and/or owner. Remedial work involving the use of curing-type compounds shall be allowed to set before it is re-checked for leakage.

8.2 Upon satisfactory reevaluation, the remedial work performed shall become punch list items to randomly check for similar conditions on the remainder of the project. When the inspection and remediation of a specimen requires removal and re-installation of parts or the whole specimen, it shall be at the sole discretion of the lab/testing agency to determine the feasibility of reevaluating the same specimen.

9.0 REPORT

9.1 The report shall include the following general information: evaluating agency, date and time of the evaluation, date of report, identification and location of the building.

9.2 The report shall include the following information regarding the fenestration specimen: Manufacturer, model, dimensions, materials, etc.; identification and locations of fenestration product(s) on the building; physical condition of fenestration product; description of any modifications made to the fenestration products.

9.3 The report shall include a statement regarding sampling procedures.

9.4 The report shall include all results (and all reevaluations) and a record of all points of water leakage.

9.5 The report shall include a compliance statement indicating that the evaluations were conducted in accordance with this method or completely describe any deviation. Also state whether the results indicate compliance with the specification requirements for the project.

Changes from AAMA 501.2-15 to AAMA 501.2-25

- Various editorial changes were made
- Added guidance for product in and out of scope
- Added requirements for testing conducted by AAMA Accredited Laboratories
- Added guidance for small section testing
- Updated Section 8 Remedial Work and Re-Checking
- This document was updated by the Quality Assurance and Water Leak Diagnostic Field Test Task Group

Changes from AAMA 501.2-09 to AAMA 501.2-15

- Updated referenced documents in Section 2.0
- Added Section 3.0 Terminology

AAMA 501.2-25

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